

[3]: #-----
print("===LIST EXAMPLE===")
fruits=["apple","banana","cherry","apple"] *#duplicates allowed*
print("Original List:",fruits)
#Add element
fruits.append("orange")
#Remove element
if "banana" in fruits:
 fruits.remove("banana")
 #Access by index
 print("First fruit:",fruits[0])
 #List comprehension
 upper_fruits=[fruit.upper() for fruit in fruits]
 print("Uppercase Fruits:",upper_fruits)

===LIST EXAMPLE===

Original List: ['apple', 'banana', 'cherry', 'apple']

First fruit: apple

Uppercase Fruits: ['APPLE', 'CHERRY', 'APPLE', 'ORANGE']

```
[6]: #-----  
print("\n===TUPLE EXAMPLE===")  
coordinates=(10.5,20.3,10.5)  
print("Tuple:",coordinates)  
print("First coordinate:",coordinates[0])
```

```
===TUPLE EXAMPLE===  
Tuple: (10.5, 20.3, 10.5)  
First coordinate: 10.5
```

```
[7]: #-----  
print("\n==TUPLE EXAMPLE==")  
coordinates=(10.5,20.3,10.5)  
print("Tuple:",coordinates)  
print("First coordinate:",coordinates[-1])
```

```
==TUPLE EXAMPLE==  
Tuple: (10.5, 20.3, 10.5)  
First coordinate: 10.5
```

```
[9]: #-----  
print("\n===TUPLE EXAMPLE===")  
coordinates=(10.5,20.3,10.5)  
print("Tuple:",coordinates)  
print("First coordinate:",coordinates[-1])  
#Tuple unpacking  
x,y,z=coordinates  
print(f"Unpacked:x={x},y={y},z={z}")
```

===TUPLE EXAMPLE===

Tuple: (10.5, 20.3, 10.5)

First coordinate: 10.5

Unpacked:x=10.5,y=20.3,z=10.5

```
[13]: def get_numbers(count=5):
    """Prompt the user to enter 'count' integers and return them as a list."""
    numbers = []
    for i in range(count):
        while True:
            try:
                num = int(input(f"Enter number {i + 1}: "))
                numbers.append(num)
                break # Exit loop if input is valid
            except ValueError:
                print("Invalid input. Please enter an integer.")
    return numbers

def main():
    # Create the list
    num_list = get_numbers()
    # Print all elements
    print("\nThe list of numbers is:", num_list)
    # Print elements one by one
    print("Elements in the list:")
    for num in num_list:
        print(num)

if __name__ == "__main__":
    main()
```

```
Enter number 1: 1
Enter number 2: 2
Enter number 3: 3
Enter number 4: 4
Enter number 5: 5
```

```
The list of numbers is: [1, 2, 3, 4, 5]
```

```
Elements in the list:
```

```
1
2
3
4
```

```
[14]: def print_first_last(lst):  
    """  
    Prints the first and last elements of a list.  
    Handles empty and single-element lists safely.  
    """  
    if not isinstance(lst, list):  
        print("Error: Input is not a list.")  
        return  
  
    if len(lst) == 0:  
        print("The list is empty.")  
    elif len(lst) == 1:  
        print(f"Only one element in the list: {lst[0]}")  
    else:  
        print(f"First element: {lst[0]}")  
        print(f>Last element: {lst[-1]}")  
  
    # Given list  
    numbers = [10, 20, 30, 40, 50]  
    # Call the function  
    print_first_last(numbers)
```

First element: 10

Last element: 50

```
[16]: # Initial list
numbers = [1, 2, 3, 4,]
# Adding 100 to the list using append()
numbers.append(100)
# Display the updated list
print("Updated list:", numbers)
```

```
Updated list: [1, 2, 3, 4, 100]
```

```
[17]: def remove_element(lst, element):  
    """  
    Removes the first occurrence of 'element' from the list if it exists.  
    Returns the updated list.  
    """  
    try:  
        lst.remove(element) # Removes by value, raises ValueError if not found  
        print(f"Element {element} removed successfully.")  
    except ValueError:  
        print(f"Element {element} not found in the list.")  
    return lst  
  
    # Initial list  
    numbers = [1, 2, 3, 4, 5]  
    print("Original list:", numbers)  
    # Remove element 3  
    numbers = remove_element(numbers, 3)  
    print("Updated list:", numbers)
```

```
Original list: [1, 2, 3, 4, 5]  
Element 3 removed successfully.  
Updated list: [1, 2, 4, 5]
```



```
[18]: def sum_of_list(numbers):  
    """  
    Returns the sum of all numeric elements in the list.  
    Raises ValueError if non-numeric values are found.  
    """  
  
    if not all(isinstance(x, (int, float)) for x in numbers):  
        raise ValueError("List must contain only numbers.")  
    return sum(numbers)  
  
# Given list  
numbers = [5, 10, 15, 20]  
  
try:  
    total = sum_of_list(numbers)  
    print(f"The sum of {numbers} is: {total}")  
except ValueError as e:  
    print(f"Error: {e}")
```

The sum of [5, 10, 15, 20] is: 50

```
[1]: # Given List
numbers = [8, 3, 9, 1, 6]

# --- Approach 1: Using built-in functions ---
max_value = max(numbers)
min_value = min(numbers)

print("Using built-in functions:")
print(f"Maximum: {max_value}")
print(f"Minimum: {min_value}")

# --- Approach 2: Manual iteration ---
# Initialize with the first element
max_val = numbers[0]
min_val = numbers[0]

# Loop through the list to find max and min
for num in numbers:
    if num > max_val:
        max_val = num
    if num < min_val:
        min_val = num

print("\nUsing manual iteration:")
print(f"Maximum: {max_val}")
print(f"Minimum: {min_val}")
```

Using built-in functions:

Maximum: 9

Minimum: 1

Using manual iteration:

Maximum: 9

Minimum: 1